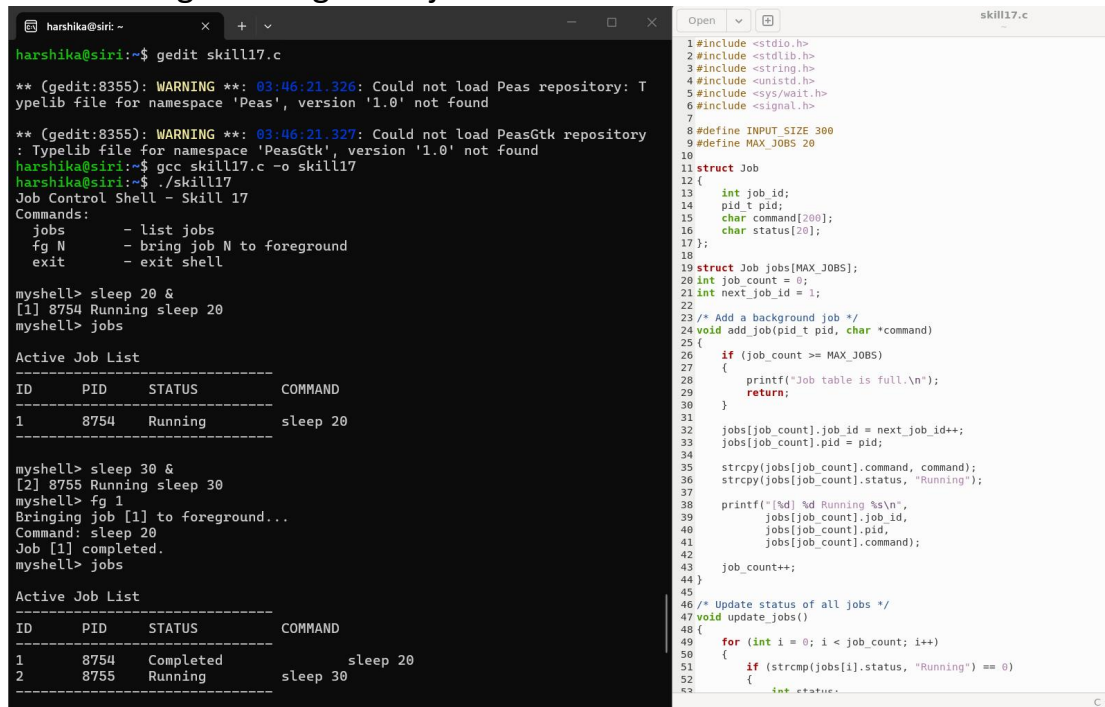


Skill - 17

1. Job listing and forground job control



The screenshot shows a terminal window with a dark background. The user is in a shell and has opened a file named `skill17.c` using `gedit`. The terminal output shows several warning messages from `gedit` about missing repositories. The user then compiles the program with `gcc skill17.c -o skill17` and runs it with `./skill17`. The program outputs a list of commands for the Job Control Shell: `jobs` (list jobs), `fg N` (bring job N to foreground), and `exit` (exit shell). The user then runs `myshell> sleep 20 &`, which outputs `[1] 8754 Running sleep 20`. The user then runs `myshell> jobs`, which displays an "Active Job List" table. The table has columns for ID, PID, STATUS, and COMMAND. It shows one job: ID 1, PID 8754, STATUS Running, COMMAND sleep 20. The user then runs `myshell> sleep 30 &`, which outputs `[2] 8755 Running sleep 30`. The user then runs `myshell> fg 1`, which outputs `Bringing job [1] to foreground...` and `Command: sleep 20`. The user then runs `myshell> jobs`, which displays an updated "Active Job List" table. The table now shows two jobs: ID 1, PID 8754, STATUS Completed, COMMAND sleep 20; and ID 2, PID 8755, STATUS Running, COMMAND sleep 30. On the right side of the screenshot, the source code for `skill17.c` is visible. It includes headers for `stdio.h`, `stdlib.h`, `string.h`, `unistd.h`, `sys/wait.h`, and `signal.h`. It defines `INPUT_SIZE` as 300 and `MAX_JOBS` as 20. The program defines a `Job` struct with fields for `job_id`, `pid`, `command`, and `status`. It then implements a `main` function that adds background jobs, updates their status, and prints the job list.

```
harshika@siri:~$ gedit skill17.c

** (gedit:8355): WARNING **: 03:46:21.326: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:8355): WARNING **: 03:46:21.327: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gcc skill17.c -o skill17
harshika@siri:~$ ./skill17
Job Control Shell - Skill 17
Commands:
  jobs      - list jobs
  fg N      - bring job N to foreground
  exit      - exit shell

myshell> sleep 20 &
[1] 8754 Running sleep 20
myshell> jobs

Active Job List
-----
ID      PID      STATUS   COMMAND
-----
1       8754     Running  sleep 20

myshell> sleep 30 &
[2] 8755 Running sleep 30
myshell> fg 1
Bringing job [1] to foreground...
Command: sleep 20
Job [1] completed.
myshell> jobs

Active Job List
-----
ID      PID      STATUS   COMMAND
-----
1       8754     Completed sleep 20
2       8755     Running  sleep 30

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4 #include <unistd.h>
5 #include <sys/wait.h>
6 #include <signal.h>
7
8 #define INPUT_SIZE 300
9 #define MAX_JOBS 20
10
11 struct Job
12 {
13     int job_id;
14     pid_t pid;
15     char command[200];
16     char status[20];
17 };
18
19 struct Job jobs[MAX_JOBS];
20 int job_count = 0;
21 int next_job_id = 1;
22
23 /* Add a background job */
24 void add_job(pid_t pid, char *command)
25 {
26     if (job_count >= MAX_JOBS)
27     {
28         printf("Job table is full.\n");
29         return;
30     }
31
32     jobs[job_count].job_id = next_job_id++;
33     jobs[job_count].pid = pid;
34
35     strcpy(jobs[job_count].command, command);
36     strcpy(jobs[job_count].status, "Running");
37
38     printf("[%d] %d Running %s\n",
39           jobs[job_count].job_id,
40           jobs[job_count].pid,
41           jobs[job_count].command);
42
43     job_count++;
44 }
45
46 /* Update status of all jobs */
47 void update_jobs()
48 {
49     for (int i = 0; i < job_count; i++)
50     {
51         if (strcmp(jobs[i].status, "Running") == 0)
52         {
53             int status;
```